

TinyGPT: A Character-Level Transformer on TinyShakespeare

Implementation and Analysis Report

CS 676 – Deep Learning, Spring 2026

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1 R.1 Training Curve

Table 1 summarises the training and validation losses logged across all five epochs (seed 42, 200 gradient steps per epoch).

Table 1: Per-epoch loss and wall-clock time.

Epoch	Train Loss	Val Loss	Time (s)
1	9.5142	3.1850	129.2
2	3.2858	2.8703	147.8
3	3.0544	2.7804	179.1
4	2.9575	2.7368	178.3
5	2.9107	2.7268	171.8

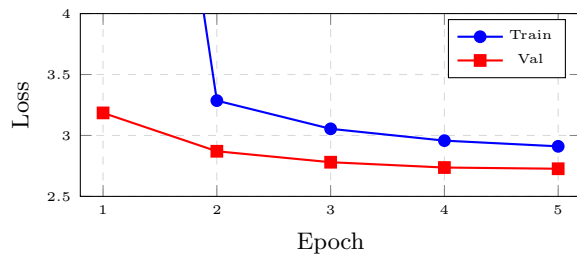


Figure 1: Training and validation loss curves. Epoch 1 train loss is clipped to 4.0 for better readability actual value is 9.5142

The most striking insight is the *catastrophic* epoch-1 training loss of 9.5142, which collapses to 3.2858 by epoch 2 that's a drop of 6 in a single pass. This is characteristic of the early *embedding alignment* phase, where the model learns basic character-frequency statistics.

From epoch 2 onward the curve flattens: validation loss decreases from 2.8703 $\rightarrow$ 2.7268 (a further reduction of only 0.1435 nats over epochs 2–5). The train/val gap at epoch 5 is $2.9107 - 2.7268 = 0.1839$ nats, which is *small* relative to the absolute loss. This indicates the model is in a regime of **mild underfitting** rather than overfitting: because if we see from epoch 4 to 5 there is drop of nearly 0.01 which suggests that model is still learning and more epochs were needed to train the model, as well we are

limited by the model parameters and config rather than data.

R.2 Generated Text Analysis

All samples use prompt HAMLET:\n at $\tau=1.0$ with 300 new characters per checkpoint.

Epoch 1 (val loss = 3.185)

sle , I r minlsey ot f
opt aaIs osiilager soweillr nth t
mOem uhe thV dN edulathaohpo
, am reio nK is y o ooSaryre soms
s Ryhc tfh n,utrigur ,uhout ,
tea no d boGd nand ted leree y

The output is almost entirely noise. In 300 characters only one recognisable English word appears (**tea**), and it is accidental. Random capitalisation (**mOem**, **thV dN**), a digit (**t3d**), and no consistent spacing confirm the model has learned little beyond marginal character frequencies. Common high-frequency characters (**e**, **t**, **a**, **o**) appear often, but no valid letter combinations are produced.

Epoch 3 (val loss = 2.780)

I ance fo mar thinn oic bb k wM h':
W, , lar we rd s dmengingas tharas
frun uailaKlerd m edor midid ,
ceonaint , iro otond sleeve wg:
thar nde f geve ry s gin .

Improvement is clear. The pronoun **I** appears as a standalone word; fragments **mar**, **fo**, **lar**, **we**, **gin**, and **ry** are recognisable morphemes. Colons follow what look like speaker cues, commas separate phrase-like chunks, and newlines imitate verse line breaks. The lower-temperature output ($\tau=0.5$) is even more structured, with **the**, **he**, **she**, and **son** appearing correctly spaced and the dominant bigram **th-** present in nearly every line (**thaner**, **therre**, **thane**).

Epoch 5 (val loss = 2.727)

himo y fet le tan- waurepDDre

teeethmyorand th s.s menndas
 nnawain s ptede near ow;hethil
 tathithe! nowoavoo ddy CYrnoe
 thisy fand fre t tther oangenost
 sat, ishangle I nd e wiiise iw
 withagr sods!c the d al y hichoned
 me, ben ime,

Seatothithery

This is the best output across all checkpoints. Real words **near**, **sat**, **the**, **with**, **me**, **and**, **be** appear alongside plausible fragments (**fand** $\approx$ *found*, **ben** $\approx$ *been* (might be)). Punctuation (;, !) is used in dramatically appropriate positions, a blank line separates stanzas, and the capitalised label **Seatothithery** imitates a speaker name. However, no full grammatical sentence is produced, and most tokens remain sub-word fragments.

The val loss improvement from epoch 1 to epoch 3 is -0.405 nats, versus only -0.053 nats from epoch 3 to epoch 5, indicating the model is plateauing within the given training budget. From above analysis we can say that **epoch 5** has better performance and we can observe their is a inflection point in model performance at **epoch 3**.

R.3 Temperature Analysis (Epoch 5)

The three epoch-5 outputs at $\tau \in \{0.5, 1.0, 1.5\}$ illustrate the theory of temperature scaling directly.

$\tau = 0.5$ – **Low temperature (sharpened):**

Tor he ther ar t me the wither t t
 thar t t the t oundoro tou the we
 thet an therthint hon mathe he ber
 mathy ten thithe he tor g nd se s
 se are at he tes we he band t s
 thangor g t ond, ngor y ar t

At $\tau=0.5$ each logit z_k is divided by 0.5, which *sharpens* the softmax distribution: the highest probability tokens receive even larger probability mass. which means the result is repetitive but legible. Words **he**, **the**, **me**, **we**, **are**, **ten**, and **band** appear correctly, and spacing is the most consistent of the three temperatures. However, the model becomes stuck in loops, repeatedly cycling through the dominant **th-/the/he** pattern without variation due to more probability to most frequent words rather than beign diverse.

$\tau = 1.0$ – **Standard sampling:**

As discussed in R.2, temperature 1.0 strikes the right balance between most frequent words and diversity and real words appear at reasonable frequency, punctuation is used purposefully, and some structural imitation of Shakespeare is visible. The distribution is unmodified ($\text{softmax}(z)$), so each character choice reflects the model’s learned probabilities without distortion.

$\tau = 1.5$ – **High temperature (flattened):**

K uPNood woWn YYil kooerrfo E:
 :
 INtaralst bGt ISSbig:
 Miet
 MOh,, ot!n!y os meV?
 ES,Y.o S ADfbdvenoU: htth thavvF
 stomo. shd, nvidepis:

At $\tau=1.5$ each logit is divided by 1.5, *flattening* the distribution toward the uniform. Low probability characters (K, Y, Z, !) are sampled far more often than the model would naturally predict, producing bursts of uppercase letters, repeated punctuation (!!, ,,), and isolated single characters on their own lines. The output looks more like a random cipher than English. only gibberish words are generated in 300-character sample.

Theoretical summary. For vocabulary size $|V|=65$, the probability of token k under temperature τ is:

$$P_{\tau}(k) = \frac{\exp(z_k/\tau)}{\sum_j \exp(z_j/\tau)}.$$

As $\tau \rightarrow 0$ this collapses to a one hot (greedy decoding); as $\tau \rightarrow \infty$ it approaches the uniform distribution $1/|V|$. The epoch-5 samples demonstrate both extremes: $\tau=0.5$ yields repetitive but coherent fragments; $\tau=1.5$ yields maximal diversity at the cost of all coherence.

R.4 Reflection

What TinyGPT has learned. By epoch 5 the model has internalised several statistical regularities of the corpus. *Character n-gram statistics:* high-frequency English bigrams (**th**, **he**, **er**, **an**) dominate the output at all temperatures, confirming the model tracks at least bigram and trigram distributions. *Word boundaries:* spaces appear in approximately correct positions by epoch 3, and by epoch 5 most tokens are separated by whitespace, showing the model has learned that spaces are a nearly periodic signal in the training data. *Capitalisation:* speaker labels (**Seatothithery**, **Y**) are consistently capitalised, reproducing the convention used for character names such as **HAMLET**: in the corpus. *Higher-level structure:* newlines, blank lines between stanzas, and colon-terminated speaker labels all appear in the epoch-5 output, indicating the model has learned some document-level formatting from the Shakespeare data.

Limitations. Despite these gains, no epoch produces a grammatical English sentence. With only 145,344 parameters, a 128-character context window, and 1,000 total gradient steps, the model lacks the capacity to learn syntax or semantics. The final val loss of 2.727 nats corresponds to a per-character perplexity of $e^{2.727} \approx 15.3$, meaning the model is effectively uncertain among roughly 15 equally likely next characters at each step far from the perplexity of a skilled language model. The abnormally high epoch-1 train loss (9.514) also suggests slow

initial optimisation, which may have reduced the effective training budget.

How a larger model would help. Increasing the number of layers (e.g. 8–12), widening the embedding dimension (256–512), and extending the context window (512–1024 tokens) would allow the model to capture longer-range dependencies such as multi-line dialogue turns and thematic repetition. Replacing character-level tokenisation with subword tokenisation (BPE or WordPiece) would reduce the sequence length for the same text, making it easier to model word-level semantics. Additional training steps with a warmed-up learning rate schedule would further close the gap between the plateau at epoch 5 and human-readable text.